



NLP-2: Practical RLHF

HSE Faculty of Computer Science

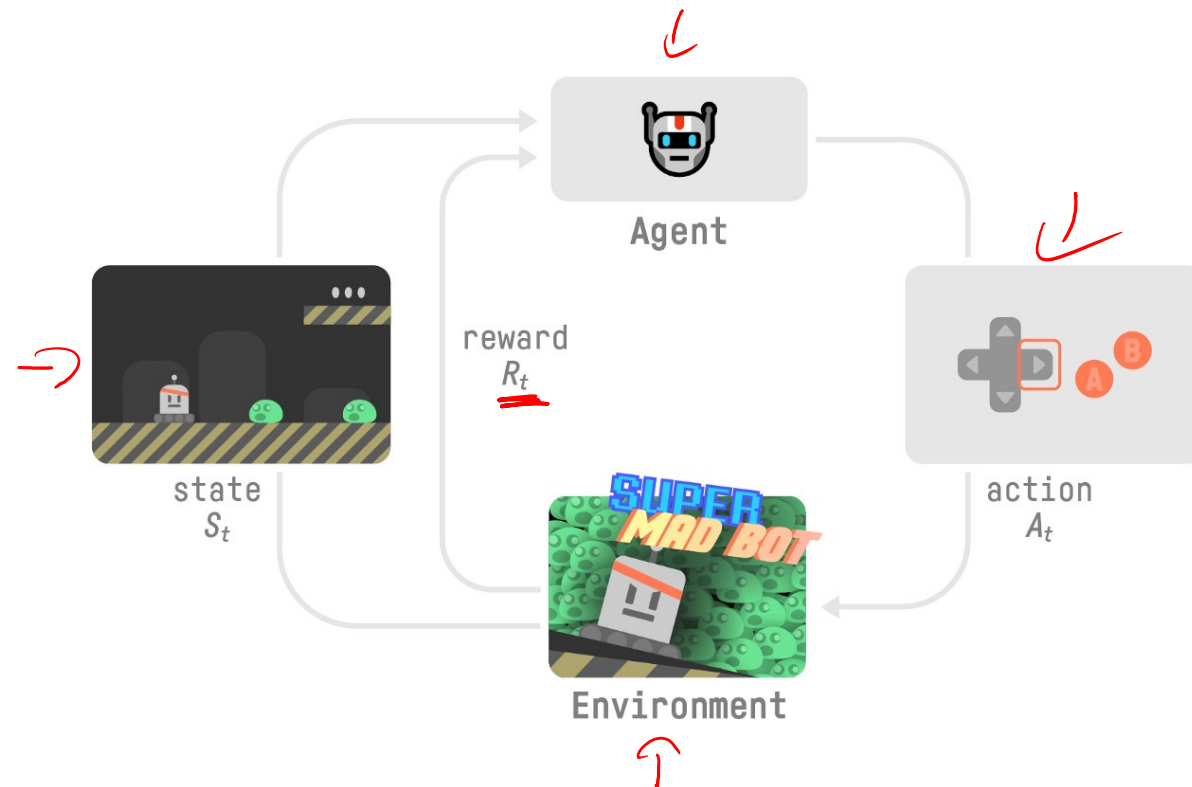
Murat Khazhgeriev



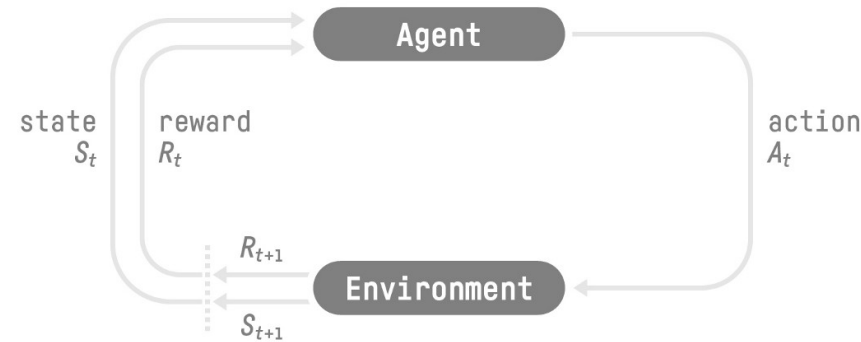
Table of Content

- **Recap**
- Actor-Critic PPO
- Introducing GRPO

An Agent acts upon an Environment, changes the State and gets a Reward



An Agent acts upon an Environment, changes the State and gets a Reward



Where to learn more?

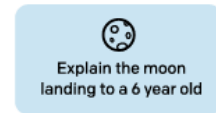
- [Deep RL Course by HF](#)
- [YSDA Practical RL](#)

RLHF is implemented in 3 main steps: SFT, RM, RL

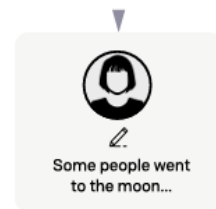
Step 1

Collect demonstration data, and train a supervised policy.

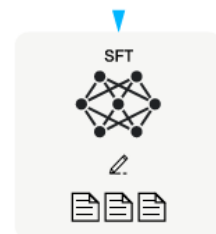
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



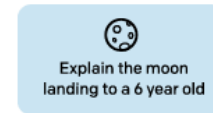
This data is used to fine-tune GPT-3 with supervised learning.



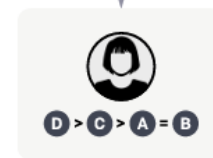
Step 2

Collect comparison data, and train a reward model.

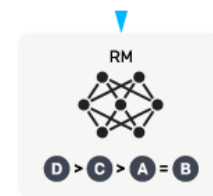
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



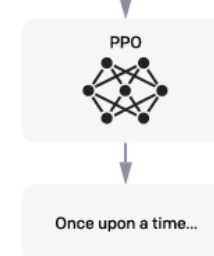
Step 3

Optimize a policy against the reward model using reinforcement learning.

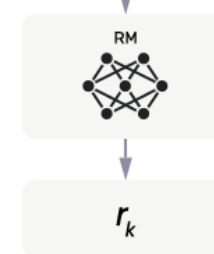
A new prompt is sampled from the dataset.



The policy generates an output.



The reward model calculates a reward for the output.



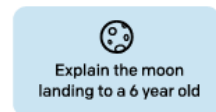
The reward is used to update the policy using PPO.

Step 1: Train an initial model with Supervised Finetuning

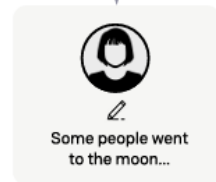
Step 1

**Collect demonstration data,
and train a supervised policy.**

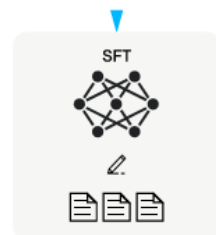
A prompt is
sampled from our
prompt dataset.



A labeler
demonstrates the
desired output
behavior.



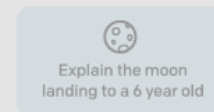
This data is used
to fine-tune GPT-3
with supervised
learning.



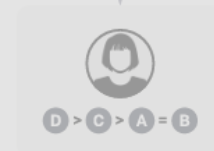
Step 2

**Collect comparison data,
and train a reward model.**

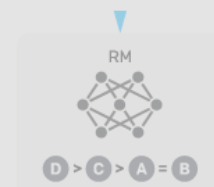
A prompt and
several model
outputs are
sampled.



A labeler ranks
the outputs from
best to worst.



This data is used
to train our
reward model.



Step 3

**Optimize a policy against
the reward model using
reinforcement learning.**

A new prompt
is sampled from
the dataset.



The policy
generates
an output.

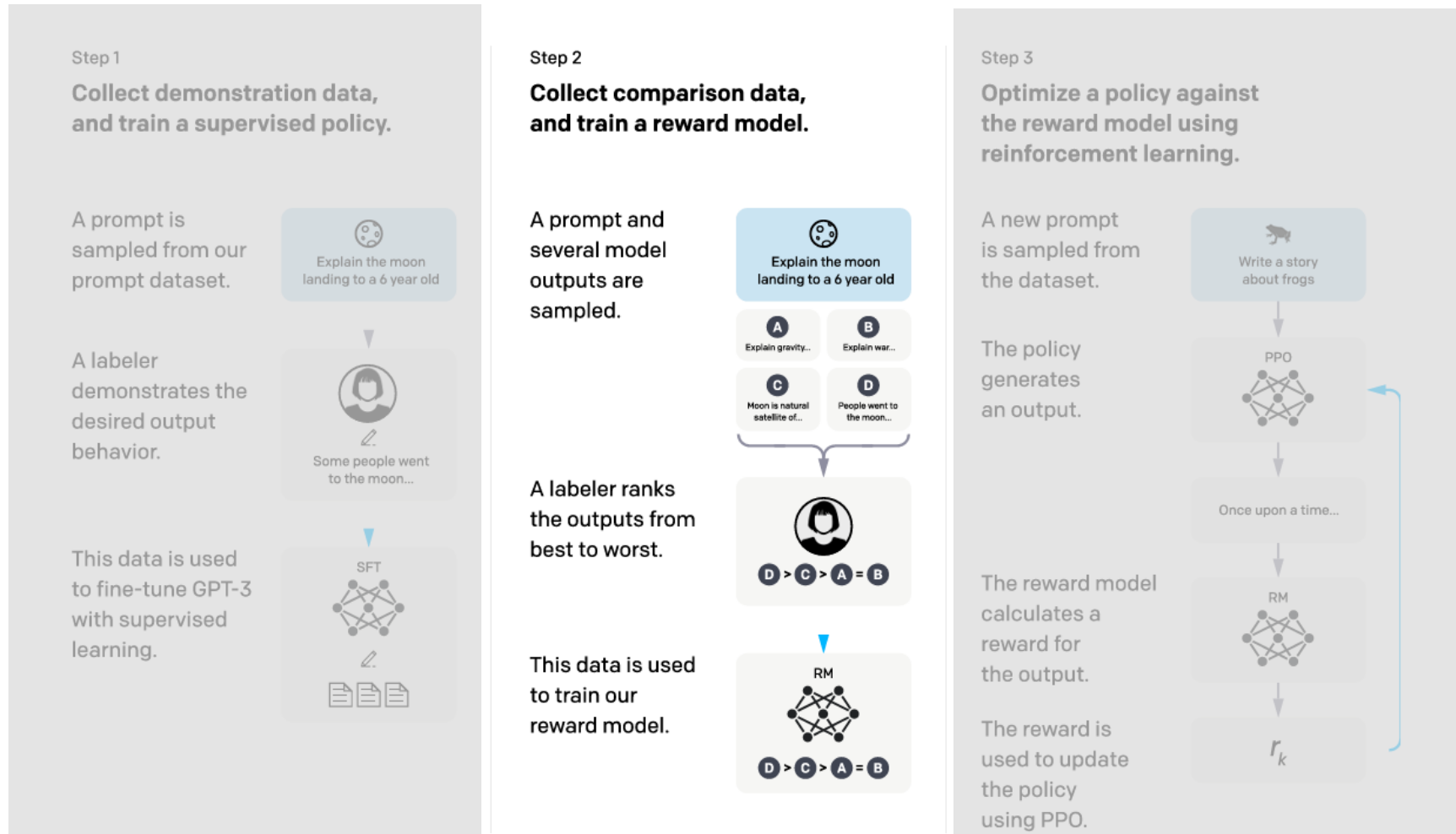


The reward model
calculates a
reward for
the output.



The reward is
used to update
the policy
using PPO.

Step 2: Train a Reward Model



Reward Model learns to differentiate between winner's and loser's scores

Bi-Encoder

$$\text{loss}(\theta) = -\frac{1}{\binom{K}{2}} E_{(x, y_w, y_l) \sim D} [\log(\sigma(r_{\theta}(\underline{x}, \underline{y_w}) - r_{\theta}(\underline{x}, \underline{y_l})))]$$

prompt completion

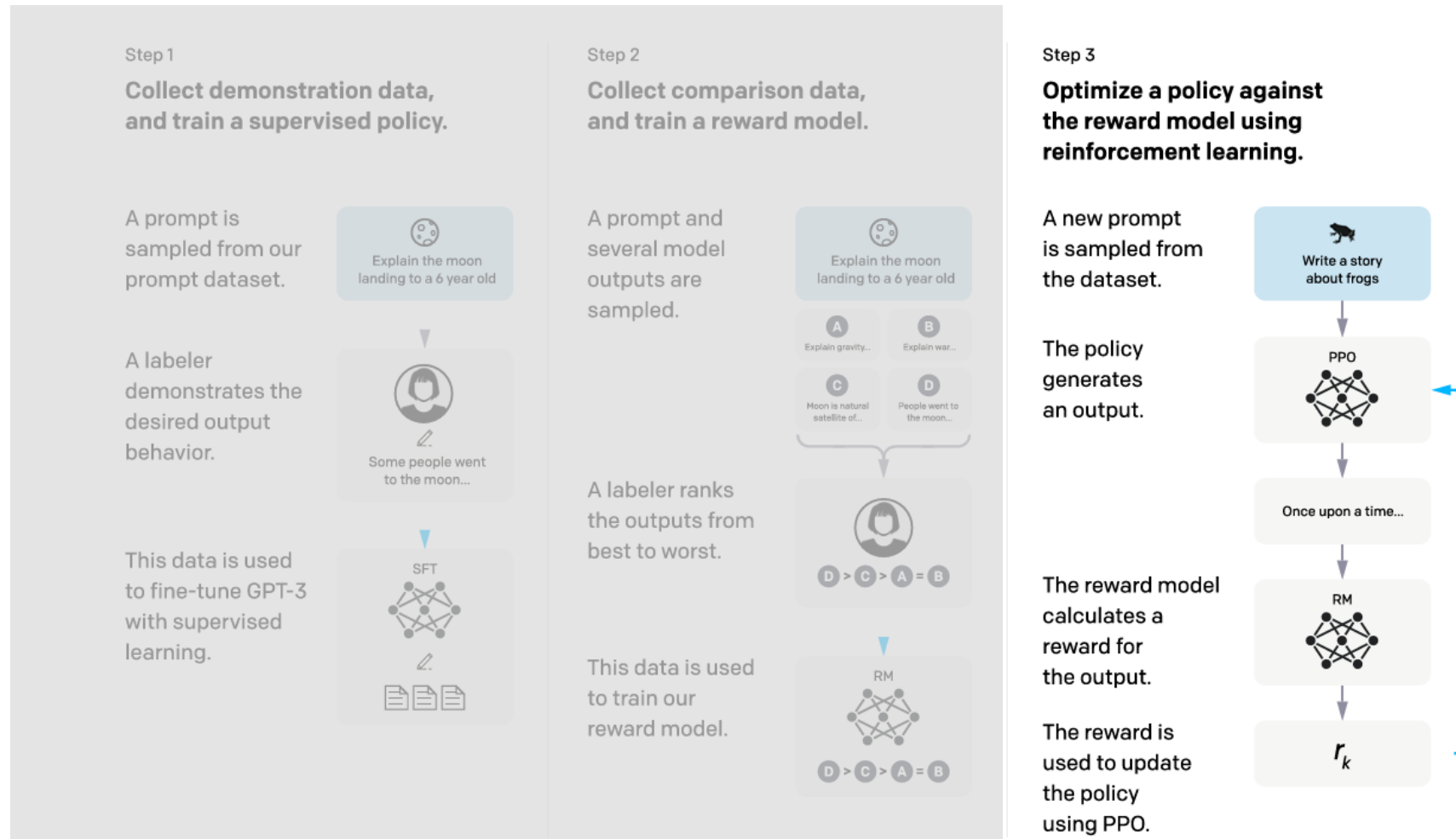
Reward Model learns to differentiate between winner's and loser's scores

$$\text{loss}(\theta) = -\frac{1}{\binom{K}{2}} E_{(x, y_w, y_l) \sim D} [\log(\sigma(r_\theta(x, y_w) - r_\theta(x, y_l)))]$$

prompt

completion

Step 3: Introduce Reinforcement Learning to the problem



Leverage a vanilla Proximal Policy Optimization setup

$$\text{objective}(\phi) = E_{(x,y) \sim D_{\pi_{\phi}^{\text{RL}}}} \left[r_{\theta}(x, y) - \beta \log \left(\frac{\pi_{\phi}^{\text{RL}}(y | x)}{\pi^{\text{SFT}}(y | x)} \right) \right]$$

KL

prompt $\rightarrow$ $\boxed{\varphi}$ $\rightarrow$ completion
" Δ $\text{text} = x, y$ $\rightarrow$ π^{SFT}
" $\times$ $\rightarrow$ π^{RL}

Your Language Model is Secretly a Reward Model

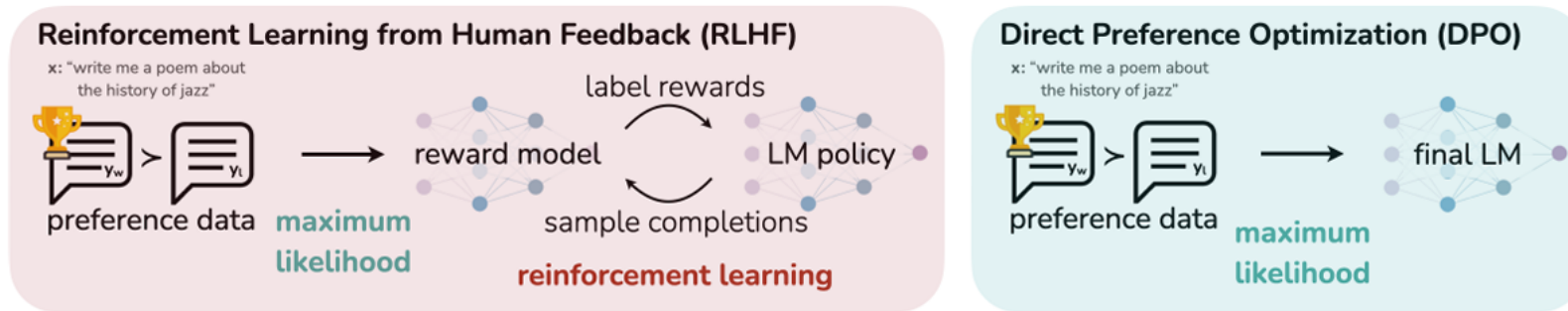


Figure 1: **DPO optimizes for human preferences while avoiding reinforcement learning.** Existing methods for fine-tuning language models with human feedback first fit a reward model to a dataset of prompts and human preferences over pairs of responses, and then use RL to find a policy that maximizes the learned reward. In contrast, DPO directly optimizes for the policy best satisfying the preferences with a simple classification objective, fitting an *implicit* reward model whose corresponding optimal policy can be extracted in closed form.

Your Language Model is Secretly a Reward Model

Reward

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\underbrace{\beta}_{\text{A}} \log \frac{\pi_{\theta}(\underline{y_w} \mid x)}{\pi_{\text{ref}}(\underline{y_w} \mid x)} - \underbrace{\beta}_{\text{B}} \log \frac{\pi_{\theta}(\underline{y_l} \mid x)}{\pi_{\text{ref}}(\underline{y_l} \mid x)} \right) \right]$$

$$\beta \downarrow \sim 0.05$$

$$\beta \uparrow \sim 1.0$$



Table of Content

- Recap
- **Actor-Critic PPO**
- Introducing GRPO

A vanilla Proximal Policy Optimization setup

$$\text{objective}(\phi) = E_{(x,y) \sim D_{\pi_{\phi}^{\text{RL}}}} \left[\underbrace{r_{\theta}(x, y) - \beta \log \left(\pi_{\phi}^{\text{RL}}(y | x) / \pi^{\text{SFT}}(y | x) \right)}_{\text{reward}} \right]$$



Rollout

$$\underline{R}_{total} = r(x, y) - \beta \log \frac{\pi^{RL}(y|x)}{\pi^{SFT}(y|x)}$$

$\begin{matrix} \times \\ \vee \end{matrix}$
prompt 0 $\rightarrow$ y_0
prompt 1 $\rightarrow$ y_1

Introducing a Critic

$x_1 \Rightarrow 9.0$

$x_2 \Rightarrow 2.5$

$$R_{total} = r(\underline{x}, \underline{y}) - \beta \log \frac{\pi^{RL}(y|x)}{\pi^{SFT}(y|x)}$$

??

$$A = R_{total} - \underline{V(x)}$$



Real PPO Loss consists of 3 parts

$$L_{PPO} = L_{policy} + c_1 L_{value} + c_2 L_{entropy}$$

Real PPO Loss consists of 3 parts

$$L_{PPO} = L_{policy} + c_1 L_{value} + c_2 L_{entropy}$$

$$L_{policy} = -\min \left(\frac{\pi_{new}}{\pi_{old}} A, \text{clip} \left(\frac{\pi_{new}}{\pi_{old}}, 1 - \epsilon, 1 + \epsilon \right) A \right)$$

Handwritten notes: $\epsilon \approx 0.2$ and $(1-\epsilon) \cdot 0, (1+\epsilon) \cdot 0$ are written above the clip function. A red circle highlights the ratio $\frac{\pi_{new}}{\pi_{old}}$ and a red arrow points to the ϵ parameter.

$$R_{total} = r(x, y) - \beta \log \frac{\pi^{RL}(y|x)}{\pi^{SFT}(y|x)}$$

$$A = R_{total} - V(x)$$

Real PPO Loss consists of 3 parts

$$L_{PPO} = L_{policy} + c_1 L_{value} + c_2 L_{entropy}$$

$$L_{policy} = -\min \left(\frac{\pi_{new}}{\pi_{old}} A, \text{clip} \left(\frac{\pi_{new}}{\pi_{old}}, 1 - \epsilon, 1 + \epsilon \right) A \right)$$

$$L_{value} = (V(x)_{pred} - R_{total})^2$$

Real PPO Loss consists of 3 parts

$$L_{PPO} = L_{policy} + c_1 L_{value} + c_2 L_{entropy}$$

$$L_{policy} = -\min \left(\frac{\pi_{new}}{\pi_{old}} A, \text{clip} \left(\frac{\pi_{new}}{\pi_{old}}, 1 - \epsilon, 1 + \epsilon \right) A \right)$$

$$L_{value} = (V(x)_{pred} - R_{total})^2$$

$$L_{entropy} = -\mathbb{E}_{s \sim \pi_{\theta_{old}}} [H(\pi_{\theta}(\cdot | s))]$$

$$H(\pi_{\theta}(\cdot | s)) = -\sum_a \pi_{\theta}(a | s) \log \pi_{\theta}(a | s)$$



Table of Content

- Recap
- Actor-Critic PPO
- Introducing GRPO



Group Relative Policy Optimization (GRPO)

$$J_{GRPO}(\theta) = \mathbb{E}_{\underline{q} \sim P(Q), \{o_i\}_{i=1}^G \sim \underline{\pi_{\theta_{old}}(q)}} \left[\frac{1}{G} \sum_{i=1}^G \left(\mathcal{L}_i^{clip} - \beta D_{KL}(\pi_{\theta} || \pi_{ref})_i \right) \right]$$

Group Relative Policy Optimization

$$J_{GRPO}(\theta) = \mathbb{E}_{q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(q)} \left[\frac{1}{G} \sum_{i=1}^G \left(\mathcal{L}_i^{clip} - \beta D_{KL}(\pi_{\theta} || \pi_{ref})_i \right) \right]$$

$$\mathcal{L}_i^{clip} = \min(\rho_i A_i, \text{clip}(\rho_i, 1 - \epsilon_{clip}, 1 + \epsilon_{clip}) A_i)$$

Group Relative Policy Optimization

$$J_{GRPO}(\theta) = \mathbb{E}_{q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(q)} \left[\frac{1}{G} \sum_{i=1}^G \left(\mathcal{L}_i^{clip} - \beta D_{KL}(\pi_{\theta} || \pi_{ref})_i \right) \right]$$

$$\mathcal{L}_i^{clip} = \min(\rho_i A_i, \text{clip}(\rho_i, 1 - \epsilon_{clip}, 1 + \epsilon_{clip}) A_i)$$

$$A_i = \frac{r_i - \text{mean}(r)}{\text{std}(r) + \epsilon}$$

$$\rho_i = \frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}$$

